

Generation Interconnection Request System Impact Study Report

For Queue Project Q12-0167

Shell Los Angeles 138 kV
February 15, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0167. The Interconnection Customer proposes a 82.0 MW solar photovoltaic facility interconnecting at the Shell Los Angeles 138 kV substation in the New Jersey area. The request was received on October 24, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 124 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,513,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Jackson Square to Tomac 138.0 kV
Contingency: Centralf to Windhub 138.0 kV
Loading: 147.6% (Rating: 289 MVA)
DFAX: 10.8%
Pre-project loading: 144.5%
2. Facility: Lumberton to Mount Laurel 138.0 kV
Contingency: Hell Creek to Windhub 138.0 kV
Loading: 145.4% (Rating: 239 MVA)
DFAX: 11.8%
Pre-project loading: 141.3%
3. Facility: Tomac to Jackson Square 138.0 kV
Contingency: Centralf to Windhub 138.0 kV
Loading: 141.6% (Rating: 301 MVA)
DFAX: 10.8%
Pre-project loading: 138.7%
4. Facility: Mount Laurel to Border Substation 138.0 kV Line
Contingency: University Wps to Centralf 138.0 kV Line
Loading: 133.3% (Rating: 233 MVA)
DFAX: 5.1%
Pre-project loading: 131.5%
5. Facility: Cartanza to Cuthbert 138.0 kV
Contingency: Dunbar to Dpc Gradiot 138.0 kV
Loading: 105.1% (Rating: 243 MVA)
DFAX: 5.7%
Pre-project loading: 103.1%

6. Facility: Lumberton to Vincent 138.0 kV
Contingency: Hell Crk to Windhub 138.0 kV
Loading: 104.4% (Rating: 361 MVA)
DFAX: 11.8%
Pre-project loading: 101.7%
7. Facility: Jackson Square to Reno 138.0 kV
Contingency: Hell Crk to Windhub 138.0 kV Line
Loading: 103.3% (Rating: 829 MVA)
DFAX: 21.6%
Pre-project loading: 101.2%
8. Facility: University Wps Substation to Mount Laurel 138.0 kV
Contingency: Greenwood to Cetralf 138.0 kV
Loading: 103.2% (Rating: 459 MVA)
DFAX: 6.3%
Pre-project loading: 102.1%
9. Facility: Centralf to University Wps 138.0 kV
Contingency: Greenwood to Centralf 138.0 kV
Loading: 102.4% (Rating: 484 MVA)
DFAX: 6.3%
Pre-project loading: 101.4%
10. Facility: Border to Seaway Sw 138.0 kV
Contingency: Windhub to Centralf 138.0 kV
Loading: 101.8% (Rating: 425 MVA)
DFAX: 7.5%
Pre-project loading: 100.4%
11. Facility: Seaway Sw to Wilmot 138.0 kV L
Contingency: Centralf to Windhub 138.0 kV line
Loading: 100.8% (Rating: 457 MVA)
DFAX: 7.5%
Pre-project loading: 99.5%
12. Facility: Wilmot to Tomac 138.0 kV
Contingency: Windhub to Centralf 138.0 kV
Loading: 100.1% (Rating: 481 MVA)
DFAX: 7.5%
Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Jackson Square to Tomac 138.0 kV	Rebuild line with higher-capacity conductor	\$926,000	2% (\$17,000)
Mount Laurel to Lumberton 138.0 kV	Rebuild line with higher-capacity conductor	\$2,507,000	3% (\$73,000)
Tomac to Jackson Square 138.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$755,000	2% (\$14,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mount Laurel to Border Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$11,208,000	1% (\$160,000)
Cartanza to Cuthbert 138.0 kV	Reconductor line	\$3,999,000	100% (\$3,999,000)
Vincent to Luberton 138.0 kV	Reconductor line	\$3,516,000	3% (\$103,000)
Jackson Square to Reno 138.0 kV	Reconductor line	\$2,682,000	2% (\$55,000)
University Wps Substation to Mount Laurel 138.0 kV	Reconductor line	\$3,066,000	1% (\$36,000)
Centralf to University Wps 138.0 kV	Reconductor line	\$297,000	1% (\$3,000)
Border to Seaway Sw 138.0 kV	Reconductor line	\$1,122,000	2% (\$23,000)
Seaway Sw to Wilmot 138.0 kV L	Reconductor line	\$466,000	2% (\$10,000)
Tomac to Wilmot Substation 138.0 kV Line	Reconductor line	\$986,000	2% (\$20,000)
	Total allocated to project		\$4,513,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.